

1. What is SQL and What are its Uses?

SQL (Structured Query Language) is a domain-specific language used to store, retrieve, manipulate, and manage data in a Relational Database Management System (RDBMS).

SQL is mainly used for handling structured data, where data is organized into tables consisting of rows and columns. SQL is generally a non-case-sensitive language.

Uses of SQL:

- SQL can retrieve data from a database.
- SQL can insert new data into a database.
- SQL can create new databases.
- SQL can create new tables in databases.
- SQL can update existing data in databases.
- SQL can delete data from databases.
- SQL can perform queries related to databases.
- SQL can create and manage database objects such as tables, views, and indexes.
- SQL can set permissions on tables, procedures, views, and other database objects.

SQL Standards:

SQL is an American National Standards Institute (ANSI) and International Organization for Standardization (ISO) standard.

Although SQL is an ANSI/ISO standard, different database management systems may have different versions and extensions of SQL. However, most of them support the major SQL commands, such as SELECT, UPDATE, DELETE, INSERT, and WHERE, in a similar manner.

2. Different Commands of SQL, Their Uses and Syntax

The following are some of the commonly used SQL commands:

1. SELECT Command Use:

The SELECT command is used to retrieve data from a database table.

Syntax:

```
SELECT column1, column2, ... FROM table_name;
```

To select all columns:

```
SELECT *FROM table_name;
```

2. UPDATE Command**Use:**

The UPDATE command is used to modify existing data in a database table. Syntax:

```
UPDATE table_name  
SET column1 = value1, column2 = value2, ... WHERE condition;
```

3. DELETE Command Use:

The DELETE command is used to delete existing records from a database table. Syntax:

```
DELETE FROM table_name WHERE condition;
```

4. INSERT INTO Command Use:

The INSERT INTO command is used to insert new records into a database table.

Syntax:

```
INSERT INTO table_name (column1, column2, column3, ...) VALUES (value1, value2, value3, ...);
```

5. CREATE DATABASE Command**Use:**

The CREATE DATABASE command is used to create a new database.

Syntax:

```
CREATE DATABASE [IF NOT EXISTS] database_name [CHARACTER SET charset_name]  
[COLLATE collation_name];
```

6. ALTER DATABASE Command**Use:**

The ALTER DATABASE command is used to modify the properties or settings of an existing database.

Syntax:

```
ALTER DATABASE [database_name] [DEFAULT] CHARACTER SET charset_name  
[DEFAULT] COLLATE collation_name;
```

7. CREATE TABLE Command Use:

The CREATE TABLE command is used to create a new table in a database.

Syntax:

```
CREATE TABLE table_name (  
column1_name data_type(size) [constraints],  
column2_name data_type(size) [constraints],  
...  
[table_constraints]  
);
```

8. ALTER TABLE Command Use:

The ALTER TABLE command is used to modify the structure of an existing table. It can be used to add, modify, or remove columns and constraints.

Syntax:

```
ALTER TABLE table_name action1,  
action2,  
...;
```

9. DROP TABLE Command

10.

Use:

The DROP TABLE command is used to permanently delete a table and all its data from a database.

Syntax:

```
DROP TABLE [IF EXISTS] table_name;
```

11. CREATE INDEX Command

Use:

The CREATE INDEX command is used to create an index on one or more columns. An index can improve the speed of data retrieval operations.

Syntax:

```
CREATE [UNIQUE | FULLTEXT | SPATIAL] INDEX index_name  
ON table_name (column1 [ASC | DESC], column2 [ASC | DESC], ...);
```

12. DROP INDEX Command Use:

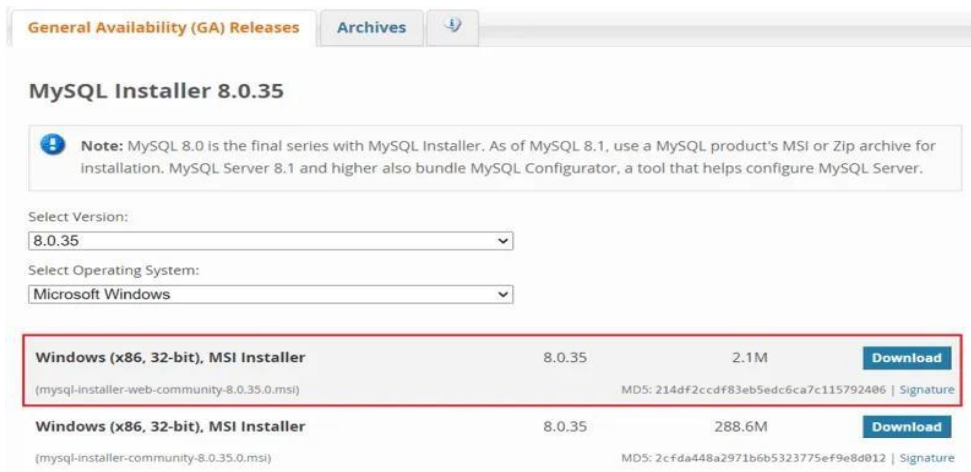
The DROP INDEX command is used to remove an existing index from a table. Syntax:

```
DROP INDEX index_name ON table_name;
```

MySQL Installation:

Step 1 – Download MySQL Installer

Select the Windows MSI Installer download.



The screenshot shows the MySQL Community Downloads page. At the top, there are tabs for "General Availability (GA) Releases", "Archives", and a download icon. Below the tabs, the title "MySQL Installer 8.0.35" is displayed. A note states: "Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server." Below the note, there are two dropdown menus: "Select Version:" with "8.0.35" selected, and "Select Operating System:" with "Microsoft Windows" selected. Below these, there is a table of download links. The first row is highlighted with a red border and contains the following information: "Windows (x86, 32-bit), MSI Installer", "8.0.35", "2.1M", and a "Download" button. Below this row, the file name "(mysql-installer-web-community-8.0.35.0.msi)" and the MD5 hash "MD5: 214df2ccd83eb5edc6ca7c115792406 | Signature" are listed. The second row contains the following information: "Windows (x86, 32-bit), MSI Installer", "8.0.35", "288.6M", and a "Download" button. Below this row, the file name "(mysql-installer-community-8.0.35.0.msi)" and the MD5 hash "MD5: 2cfd448a2971b6b5323775ef9e8d012 | Signature" are listed.

Download Link	Version	Size	Action
Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.35.0.msi)	8.0.35	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.35.0.msi)	8.0.35	288.6M	Download

Step 2 – Start the Download

Select “No thanks, just start my download.”



The screenshot shows the MySQL Community Downloads page. At the top, there is a heading "MySQL Community Downloads" with a download icon. Below the heading, there is a section titled "Login Now or Sign Up for a free account." with the text "An Oracle Web Account provides you with the following advantages:" followed by a list of benefits: "Fast access to MySQL software downloads", "Download technical White Papers and Presentations", "Post messages in the MySQL Discussion Forums", and "Report and track bugs in the MySQL bug system". Below the list, there are two buttons: "Login » using my Oracle Web account" and "Sign Up » for an Oracle Web account". Below the buttons, there is a paragraph of text: "MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can signup for a free account by clicking the Sign Up link and following the instructions." At the bottom of the page, there is a button labeled "No thanks, just start my download." which is highlighted with a red border.

No thanks, just start my download.

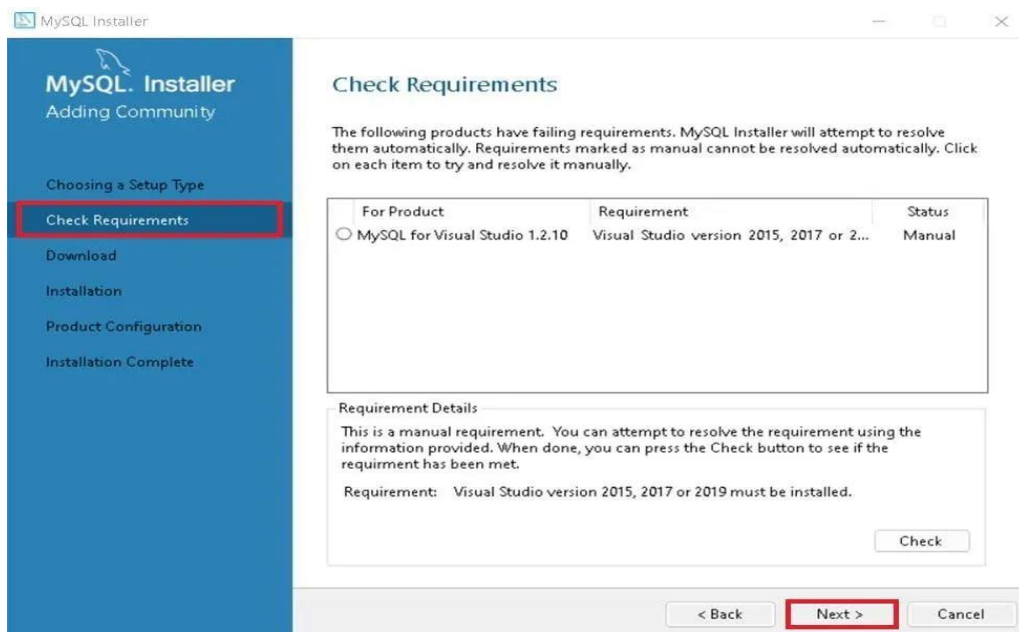
Step 3 – Wait for the Installer

Wait while Windows configures MySQL Installer.



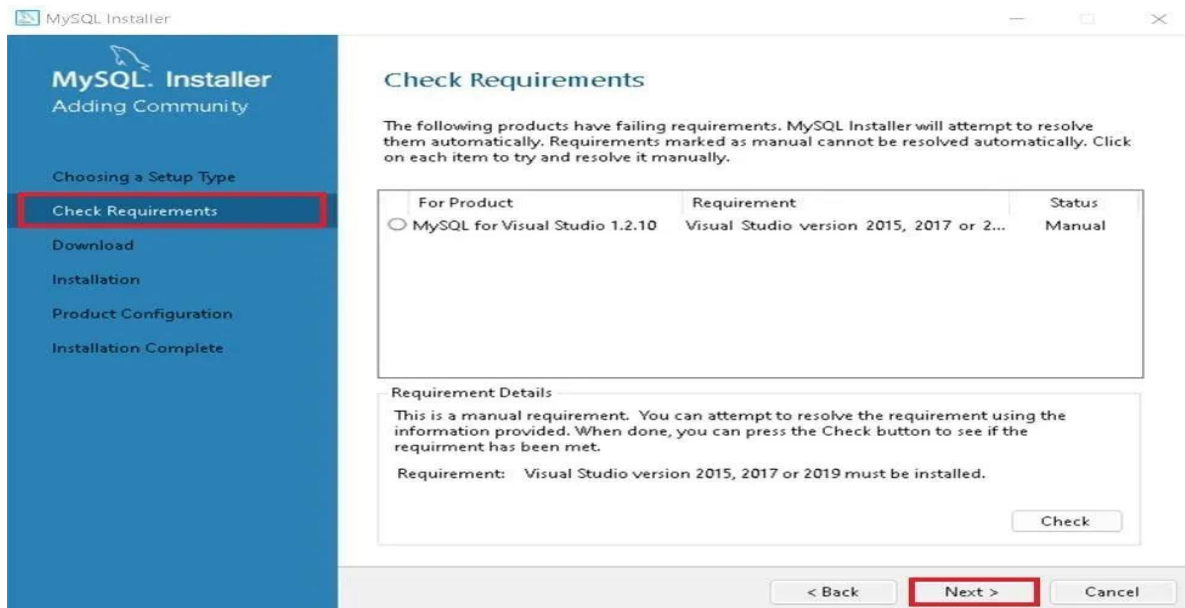
Step 4 – Check Requirements

Check the listed requirements and continue with Next.



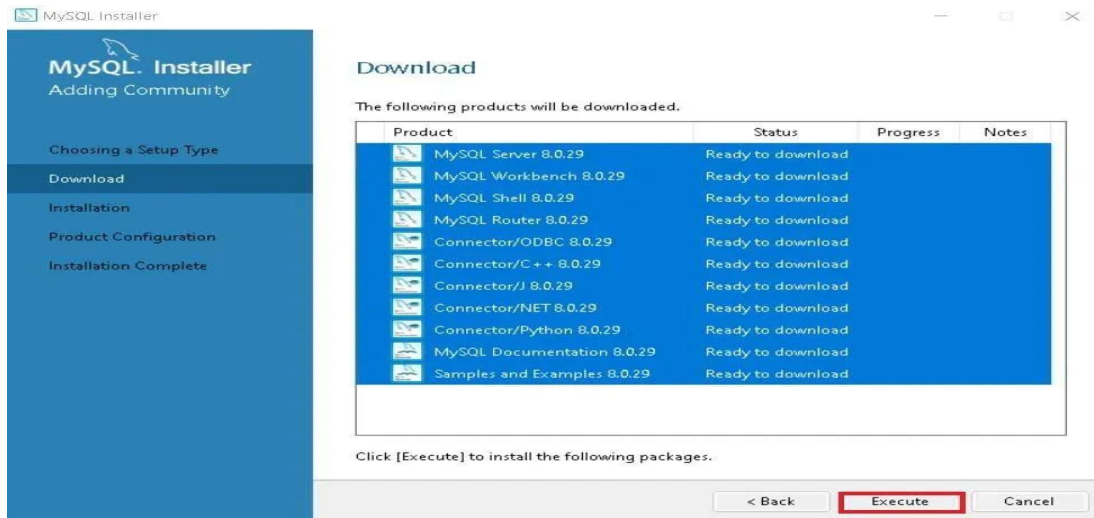
Step 5 – Continue

Continue to the Download section.



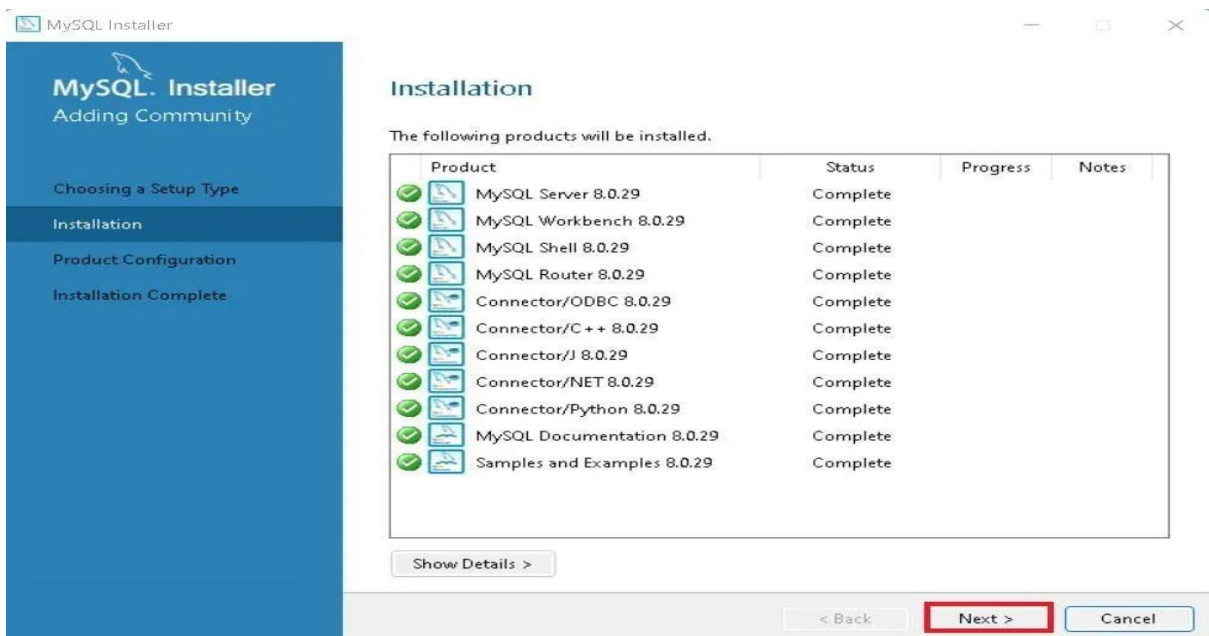
Step 6 – Download Packages

Click Execute to download and install the listed packages.



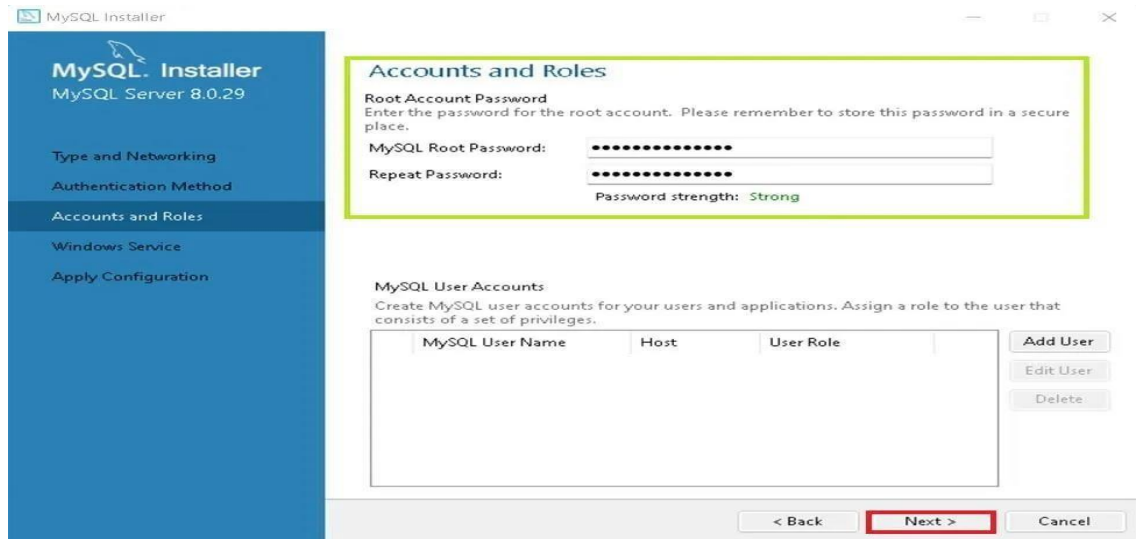
Step 7 – Installation

Wait until the listed products show Complete, then click Next.



Step 8 – Accounts and Roles

Enter the MySQL Root Password and repeat it.



The screenshot shows the 'Accounts and Roles' step of the MySQL Installer. The left sidebar lists the installation steps: Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter the password for the root account and repeat it. The password fields are filled with dots, and the password strength is indicated as 'Strong'. The second section, 'MySQL User Accounts', prompts the user to create MySQL user accounts. It features a table with columns for MySQL User Name, Host, and User Role. To the right of the table are buttons for 'Add User', 'Edit User', and 'Delete'. At the bottom of the window are navigation buttons: '< Back', 'Next >' (highlighted with a red box), and 'Cancel'.

MySQL Installer

MySQL. Installer
MySQL Server 8.0.29

Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: Strong

MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

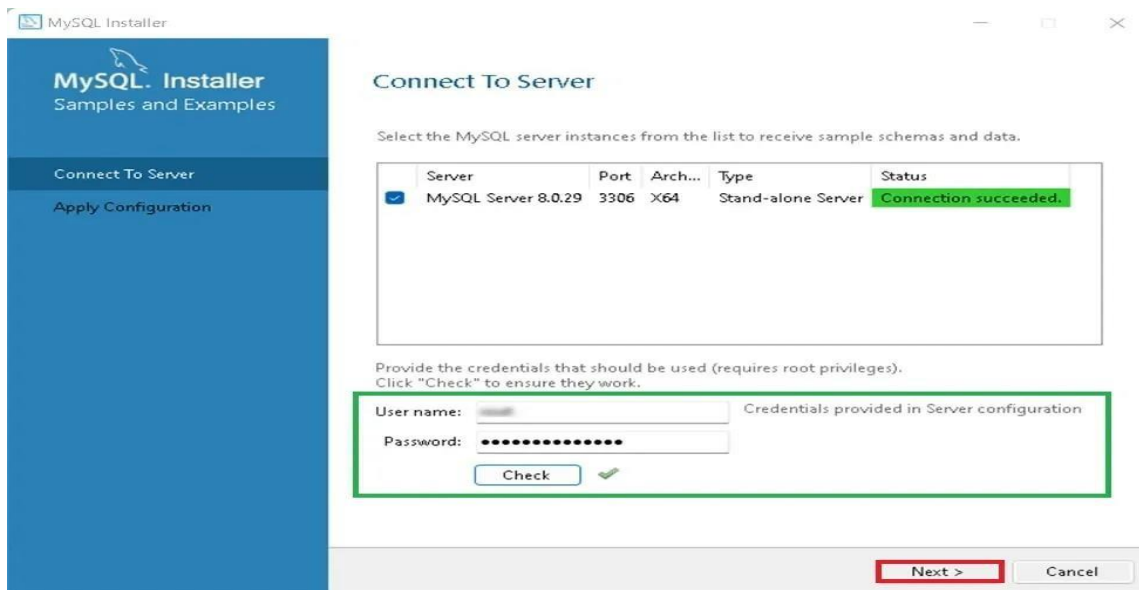
MySQL User Name	Host	User Role
-----------------	------	-----------

Add User
Edit User
Delete

< Back **Next >** Cancel

Step 9 – Connect to Server

Enter the credentials and click Check until the connection succeeds.



The screenshot shows the 'Connect To Server' step of the MySQL Installer. The left sidebar lists the installation steps: Connect To Server (selected) and Apply Configuration. The main area is titled 'Connect To Server' and contains two sections. The first section prompts the user to select the MySQL server instances from the list to receive sample schemas and data. It features a table with columns for Server, Port, Arch..., Type, and Status. The first row, 'MySQL Server 8.0.29', is selected with a checkmark, and its status is 'Connection succeeded'. The second section prompts the user to provide the credentials that should be used (requires root privileges). It features input fields for 'User name' and 'Password'. The 'User name' field is filled with 'root', and the 'Password' field is filled with dots. To the right of the input fields is the text 'Credentials provided in Server configuration'. Below the input fields is a 'Check' button with a green checkmark icon. At the bottom of the window are navigation buttons: 'Next >' (highlighted with a red box) and 'Cancel'.

MySQL Installer

MySQL. Installer
Samples and Examples

Connect To Server
Apply Configuration

Connect To Server

Select the MySQL server instances from the list to receive sample schemas and data.

Server	Port	Arch...	Type	Status
☒ MySQL Server 8.0.29	3306	X64	Stand-alone Server	Connection succeeded.

Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.

User name: root Credentials provided in Server configuration

Password:

Check

Next > Cancel

Step 10 – Installation Complete

Click Finish to complete the installation

